УДК 004.94

АВТОНОМНЫЕ КОСМИЧЕСКИЕ СИСТЕМЫ ОСНОВАННЫЕ НА ИСКУССТВЕННОМ ИНТЕЛЛЕКТЕ.

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В данной статье рассмотрена перспектива развития новейшей информационной технологии-нейросети. Данная технология может внести существенный вклад в становлении новой ступени научно-технического прогресса.

Ключевые слова: нейросетевые технологии, космическое пространство, постинформационное общество, научно-технический прогресс, нейросеть.

AUTONOMOUS SPACE SYSTEMS BASED ON ARTIFICIAL INTELLIGENCE.

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This article considers the prospect of developing the latest information technology-neural network. This technology can make a significant contribution to the formation of a new stage of scientific and technological progress.

Keywords: neural network technologies, outer space, post-information society, scientific and technological progress, finance

This study shows that the rapid development of many neural network technologies gives rise to active space exploration, determines the main trends in the development of modern society. Neural network technologies make it possible to expand the knowledge base for the introduction of newer technologies, thereby complicating the development processes in the field of autonomous space vehicles (ASVs) for space exploration, which in turn leads to new problems in the use of ASCs.

AKA, based on neural network technologies, aims to explore outer space through the creation of terrain maps, the sampling of the surface of planets and celestial bodies, the study of space climate, and weightlessness. These processes can be implemented in different ways, but the most important parameter for choosing a method will be the effective interaction of all necessary research processes using neural networks for further implementation of the prospects for using ACA [1].

The main prospects for the use of neural network technologies for the work of the ASC are: automation of the process of exploring outer space for the purpose of its further colonization, solving complex problems of communications in outer space, studying the conditions of outer space to develop measures to ensure human life.

To realize these prospects, it is necessary to organize comprehensive solutions to eliminate possible problems in the process of space exploration, which are the following factors:

1. The electrification of the spacecraft during high speed modes, which creates a fire hazard;
2. The problem of operational identification of the current state of on-board systems due to a break in communication;
3. Irradiation of spacecraft with solar radiation, which leads to the failure of the neural network complex;
4. Training of neural networks for recognition of space objects [2].

The use of ACA on a neural network basis allows you to explore outer space to create new technologies that will be used on our planet. So, for example, solar panels were originally created to ensure the autonomy of the spacecraft, and eventually led to the emergence of solar energy, which in turn can lead to an energy revolution, as shale oil did in its time. Let's take the Hubble Space Telescope as an example. Hubble is an automatic observatory in Earth orbit, which was created solely for the purpose of scientific research. Thanks to this automatic observatory, huge amounts of information about stars and galaxies were obtained, which further required the creation and implementation of technologies for transmitting information from space to Earth, which was the impetus for the development of satellite systems, which are used, for example, in GSM and GPRS technologies (cellular communications) [3]. Thus, we can conclude that the use of neural network-based ASC for space exploration is one of the investment tools that, in the long run, can pay back the investment many times over by introducing new technologies to the masses [4].

As already mentioned earlier, various methods can be used for space exploration, for example: the use of orbital stations, space probes, manned flights, the use of landing modules. All these methods imply the direct presence of a person, which in turn creates a risk to his life. Thus, one of the main advantages of using AKA on a neural network basis is its autonomy and the ability to control in a mediocre way with the help of control rooms. You can also highlight a number of the following important advantages of using ACA on a neural network basis:

1. Safe automated study of outer space conditions;
2. Fast transfer of research data;
3. The ability to study the resources of planets and asteroids to develop ways to use and further prepare for their collection;
4. Full autonomy of work on solar energy;
5. Fast and efficient analysis of huge arrays of outer space;

Thus, in the circumstances of the rapid formation of the current post-information society, in which there is a continuous and significant increase in demand for the use of the latest information technologies, one of the most basic nuances is the creation of an automated concept for managing these technologies, the basis of which will include a neural network. This technology itself will take us to the latest stage of scientific and technological progress, because thanks to it the latest technologies will be introduced everywhere, due to which there will be a rapid increase in financial performance, as this skill, according to research, will be adopted in various areas of society [5-8].

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